# Sonic AI — Assignment Architecture & Engineering Explanation

AcmeCorp Autonomous Collections Agent & Deterministic Bank Reconciliation Engine

## 1. System Purpose & Core Objective

The Sonic AI platform automates accounts receivable collections and bank statement reconciliation for AcmeCorp India Pvt Ltd. B2B receivables involve sensitive commercial contracts, tax deductions (TDS), and dispute risks. Consequently, Sonic AI implements a strict **governed human-in-the-loop** architecture. The system calculates dynamic overdue schedules, drafts context-isolated email communications, enforces factual verification against database ground truth, and deterministically scores bank receipts without delegating financial commitments or balance updates to probabilistic LLM decisions.

## 2. Collections Agent & Customer Reply Intelligence

**A. Dynamic Overdue Determination:** Invoices are never statically marked overdue. An invoice is overdue iff status == 'sent', balance_due > 0, and due_date < reference_date. Draft and already-paid invoices are strictly barred from reminder queues.
**B. Customer Isolation & Minimum Context:** When classifying inbound messages, LLM prompts receive *only* the verified invoices of that specific customer and official corporate policies. Full customer databases and foreign invoices are never exposed.
**C. Ground Truth Fact Validation:** Every drafted invoice number, due date, and monetary amount is cross-referenced against MongoDB before dispatch. Hallucinated invoice references or unauthorized payment extension commitments are intercepted and routed to the Needs Attention queue.
**D. Golden Rule Governance:** A customer's promise or claim of payment does *never* mark an invoice paid. Invoices are only cleared upon explicit human review and verified bank transaction reconciliation.

| Category | Escalation Routing | System Behavior |
|---|---|---|
| routine_query | Safe Auto-Reply / Draft | Returns verified balance, open invoices, and HDFC Bank coordinates. |
| payment_promise | Log Promise | Records promised clearance date; keeps invoice balance open. |
| payment_claim | Acknowledge Claim | Acknowledges UTR/claim; defers paid status to bank reconciliation. |
| extension_request | Needs Attention (Human) | Accounts Team approval strictly required; AI makes no commitment. |
| dispute | Needs Attention (Human) | Enforces AcmeCorp 3-day SLA; assigns finance operations desk. |
| legal_threat | Critical Priority Review | Immediate legal escalation; blocks any automated communication. |
| unknown_sender | Needs Attention (Human) | Zero disclosure of sensitive financial information; customer ID check. |

## 3. Deterministic Bank Reconciliation Engine

The reconciliation engine matches imported bank statement rows against accounts receivable invoices using a deterministic 5-signal scoring algorithm calibrated with RapidFuzz string normalization. Raw bank statements are never sent to LLMs, ensuring 100% deterministic, audit-proof matching.

| Signal | Weight | Engineering Implementation & Heuristics |
|---|---|---|
| Invoice Number / Ref | 30% | Regex extraction for INV-XXXX, RE-XXXX, numeric tokens. Exact match provides strong evidence. |
| Payer Name Similarity | 30% | Strips noise terms (PVT LTD, CORP, INC, RTGS, DUES). RapidFuzz token_set_ratio handles reordering. |
| Amount Similarity | 25% | Detects exact match, valid partial receipt (< balance), and domestic 10% TDS deduction (subtotal * 0.90 + tax). |
| Date Proximity | 10% | Compares transaction date with invoice due date (<=15d: 1.0, <=45d: 0.8, <=90d: 0.5). |
| Currency Match | 5% | Hard reject: currency mismatch immediately zeroes the confidence score to prevent cross-currency errors. |

**Debit Row & Duplicate Exclusion:** Debit entries (office rent, software vendor) are automatically identified and excluded from receipt reconciliation. Deduplication logic prevents duplicate bank transaction references from creating duplicate matches.
**Human Approval & Idempotency:** Payments are *never* applied automatically. A human operator chooses whether to close an invoice with 10% TDS deduction (upon Form 16A receipt) or leave the balance open for partial payments. Duplicate approval clicks are strictly idempotent and cannot double-apply funds.

## 4. Technical Stack & Deployment Integrity

• **Frontend:** React 18 + TypeScript + Vite + Tailwind CSS with modern B2B fintech UI (Linear/Stripe inspired).
• **Backend:** Python FastAPI with Pydantic v2 schemas and modular service-repository architecture.
• **Database:** PyMongo connected to MongoDB Atlas / local MongoDB, with automatic fallback to high-fidelity in-memory mongomock for zero-setup evaluation.
• **AI / LLM:** Google Gemini API with fallback rule engine when running offline or without credentials.
• **Auditing:** Complete audit logging tracks every customer update, email send, and reconciliation approval.